Modeling Glioblastoma Tumor Progression using Machine Learning Analysis of Longitudinal MRI Data

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ABSTRACT– Glioblastoma (GBM) is an aggressive primary brain tumor with rapid progression and limited survival, making accurate monitoring of tumor dynamics essential for clinical decision-making. While MRI is the standard modality for assessing GBM progression, most available datasets provide only single-timepoint imaging, hindering efforts to model tumor evolution. The LUMIERE dataset addresses this limitation by providing longitudinal MRI data across multiple postoperative timepoints. Using this dataset, we developed machine learning models to predict patient-specific tumor volume changes following surgical resection. Tumor volumes were extracted from T1-weighted MRI and total tumor volume was computed from enhancing, non-enhancing, and edema compartments. Temporal features including prior tumor volumes, irregular imaging intervals, and logarithmic growth ratios were engineered to capture tumor trajectory and instantaneous growth rate. Three regression models were trained using patient-level splits: Random Forest, XGBoost, and a neural network. Random Forest and XGBoost demonstrated strong predictive accuracy (R^2 = 0.84–0.88), while the neural network performed moderately (R^2 = 0.70). These findings show that temporally informed machine learning models can effectively capture GBM progrbession and may support future clinical forecasting applications with additions of radiomic and clinical features.

I. INTRODUCTION

Glioblastoma (GBM) is the most aggressive and most frequently diagnosed malignant primary brain tumor in adults. GBM grows rapidly, infiltrates surrounding healthy tissue, and almost always recurs even after intensive therapy. Standard treatment typically includes maximal safe surgical resection followed by radiation therapy and temozolomide chemotherapy, yet median survival remains only 15 to 20 months. Long-term survival is extremely rare. Because GBM changes quickly and unpredictably, clinicians rely heavily on magnetic resonance imaging (MRI) to monitor tumor progression, evaluate treatment response, and guide clinical decision-making. MRI provides several complementary contrasts, including pre- and post-contrast T1-weighted imaging, T2-weighted imaging, and FLAIR. These modalities reveal different aspects of tumor biology such as edema, necrosis, and contrast-enhancing tumor. The

Response Assessment in Neuro-Oncology (RANO) criteria use these contrasts to define progression and guide treatment assessments [4].

Despite MRI's key role, most publicly available GBM datasets contain only a single imaging timepoint per patient. Popular datasets like BraTS have greatly advanced segmentation research but focus primarily on preoperative or early postoperative scans. This prevents meaningful study of how tumors change over time. Meanwhile, emerging datasets such as the gbm-data-longitudinal repository [2], the UCSF longitudinal GBM cohort [3], and the UCSD PT-GBM dataset [7] show increasing interest in capturing tumor evolution, yet these efforts are still limited in size, consistency, and completeness. Due to the lack of robust longitudinal data, most prior work centers on single-timepoint tasks such as segmentation, radiomics-based feature extraction, molecular subtype prediction, and survival analysis [1], [6]. As a result, the field lacks strong models that describe or predict tumor behavior across the disease course.

RELATED WORKS

Several research directions have attempted to address GBM progression, but each has notable limitations. Mechanistic tumor growth models, including reaction–diffusion and proliferation-invasion frameworks, attempt to simulate tumor expansion mathematically. While these models are theoretically grounded, they require individualized parameter fitting, rely on simplifying biological assumptions, and are rarely validated across large patient cohorts [9].

Deep learning and radiomics-based approaches have achieved strong performance in tasks such as survival prediction, MGMT status classification, and tumor segmentation. However, these methods almost always rely on single MRI timepoints due to data constraints, preventing them from capturing patient-specific tumor trajectories [5], [6]. A small number of studies have used handcrafted features or limited longitudinal datasets to predict tumor progression, but these models commonly face challenges such as variable scan intervals, small sample sizes, and missing clinical annotations [8]. No widely used framework currently exists that models GBM using repeated MRI scans across multiple disease stages.

The lack of consistent longitudinal datasets has therefore created a major gap in neuro-oncology: despite MRI being the central tool for GBM management, research models rarely incorporate the temporal dimension of tumor growth.

STUDY GOAL AND CONTRIBUTION

The LUMIERE dataset addresses many of the foregoing limitations by providing 638 imaging sessions from 91 GBM patients, totaling nearly 2,500 MRI images across multiple modalities [1]. Each patient has multiple scans spanning pre-treatment and post-treatment stages, along with clinical and molecular information including

MGMT promoter methylation, IDH1 mutation status, RANO progression assessments, and survival outcomes. This dataset offers a rare opportunity to analyze GBM as a dynamic, evolving disease rather than as isolated static images.

Building on existing work that links imaging features to tumor biology, this study trains machine learning (ML) models– Random Forest, Extreme Gradient Boosting (XGBoost), and a Sequential Neural Network–to predict future tumor volume from earlier MRI-derived features. Decision Tree models are well suited for this task because they handle nonlinear relationships, perform reliably on smaller datasets, and resist overfitting in high-noise medical data environments [5]. While such models have been used widely in radiomics and oncology, they have rarely been applied to longitudinal GBM modeling and have not been evaluated at the scale or temporal depth offered by LUMIERE [8], [10], [15].

The main contributions of this study are:

1. A longitudinal modeling framework that uses repeated MRI scans to predict future tumor progression.
2. The first systematic evaluation of machine learning models for multi-timepoint GBM prediction using the LUMIERE dataset.
3. Evidence that earlier MRI features can estimate future tumor volume, supporting movement toward dynamic and personalized modeling.
4. A demonstration of the practical value of longitudinal machine learning, which may assist future treatment planning and disease monitoring.

By shifting from static imaging analysis to temporal modeling, this work helps advance the field toward tools that can better characterize how GBM evolves and potentially support more informed clinical decisions.

II. Methods

We use regression models to determine the best performing ML model to predict patient-specific GBM tumor size (volume) changes over time across multiple post-operative imaging time points following the surgical resection of the tumor.

DATASET AND PREPARATION

As aforementioned, the LUMIERE dataset contains 91 GBM patients, totalling to about 2,147 MRI images; however, not all patients or images were used. The data was extracted from the database and compiled into a .csv file to allow for simpler manipulation of data– including tumor volumes (enhancing core, non-enhancing, edema compartment) and timepoints (weeks). The voxel volumes were extracted using python library, nibable, a well-known library used for MRI image analysis including reading, writing, and accessing the NIfTI file format. Following data extraction and computation of voxel size (mm^3) for each image, enhancing, non-enhancing, and edema compartment volumes were computed. From these volumes, total tumor volume (cm^3) was calculated to provide the

model a target for prediction. The total tumor volume was scaled down to cm^3 to provide the model with numbers of lesser magnitude to reduce probability of poor model performance.

In the LUMIERE database, each patient's dataset contains initial time points which are denoted as "week-000-1" and "week-000-2." These time points are known as the pre-operative time points where the original tumor mass was still present in the patient's brain. These points were excluded from our working dataset as they are outside the scope of our focus. The processed dataset only represents data points after these two timepoints to observe and predict how the tumor grows post-surgical resection. Before building the model and engineering features, patients with less than 4 weeks (data points) were dropped from the final dataframe. This was to increase model accuracy as timepoints would perform poorly based on the nature of the engineered features. The final dataset contains 50 patients with 4 or more data points to conduct ML modeling and prediction.

MODEL ARCHITECTURE

This study incorporates three linear regression models to create predictions for total tumor volume based on a training test split of patients using the processed dataset. The models used in this study are RandomForest, XGBoost, and keras sequential neural network model. These were chosen based on their ability to learn longitudinal and create models to predict the target variable.

Random Forests Prediction Model

A Random Forest regressor was implemented to predict continuous tumor volume changes over time. The random forest regression model is a supervised learning technique which predicts a continuous target variable given input variables. It does this by averaging the predictions of multiple decision trees. Thus, creating a forest of decision trees using randomly selected subsets of the training data, features, and groups. The model used NumPy for numerical computation, Pandas for managing tumor-volume data, and scikit-learn for label encoding, k-fold grouping, dataset split, permutations, and the random forest algorithm.

XGBoost Prediction Model

An XGBoost model was implemented to predict the continuous tumor volume changes over time. It is an optimized gradient boosting library designed to be highly efficient, flexible, and portable. It is an ensemble technique that gives more weight to misclassified predictions in previous models to enhance overall model performance. It combines weak model predictions into one ensemble. These are then used as decision trees that minimize errors to increase overall accuracy. Similar to Random Forests, this combines multiple decision tree regressors to capture many complex relationships between features. The model used NumPy for numerical computation, Pandas for managing tumor-volume data, and scikit-learn for

label encoding, k-fold grouping, dataset split, permutations, and the XGBRegressor algorithm.

Neural Network

A Neural Network was constructed to assess its performance on predicting tumor volume changes over time. A machine learning model based on creating neurons in multiple layers was created to compare its performance to random forests and XGBoost. The model is a Sequential keras model consisting of an input layer (7 inputs), three hidden layers (128, 64, 64, neurons respectively) and an output layer with one neuron (real predicted value). The dense hidden layers use activation function 'relu' and the output layer with no activation function. The model was compiled using 'nadam'(optimizer), mean squared error (MSE) (loss), and mean absolute error (MAE) (metrics). As with the decision-tree models, It was then trained and fit exclusively with the training data using 200 epochs and a 0.2 validation split.

The models' independent variables (features, X) included: week number, Lag_1, Lag_2, dWeek_1, dWeek_2, log_ratio_1, and log_ratio_2. The dependent variable (target, y) was the total tumor volume (mm^3).

FEATURE SELECTION + ENGINEERING

Week number was used as the primary temporal variable in the model, providing a chronological reference point for all imaging measurements. This feature establishes the temporal axis along which tumor progression is evaluated and anchors the interpretation of all subsequent time-dependent variables. The lag features (Lag_1 and Lag_2) represent tumor volumes from one and two preceding time points, respectively, enabling the model to infer short-term and intermediate historical trends. These features allow the model to incorporate memory of prior tumor states, which is essential for capturing momentum and directionality in tumor growth.

Because patients are imaged at irregular intervals, dWeek_1 and dWeek_2 were included to quantify the elapsed time between consecutive measurements. These variables correct for variability in follow-up schedules and ensure that changes in tumor size are interpreted relative to the amount of time that has passed.

To capture the intrinsic dynamics of tumor progression, logarithmic volume ratios were computed between successive time points. These features quantify proportional, rather than absolute, changes in tumor size and serve as estimates of instantaneous growth rate. Positive values indicate tumor enlargement, negative values indicate shrinkage, and values near zero correspond to stability. The logarithmic transformation is particularly appropriate in oncology because tumor growth often follows multiplicative or exponential kinetics. By linearizing these nonlinear patterns, the model can more consistently learn growth behavior across patients with widely varying baseline tumor sizes.

$$\log\left(\frac{V_t}{V_{t-1}}\right)$$

Equation 1. Log ratio equation to determine instantaneous growth rate

FEATURE IMPORTANCE

Feature importance was evaluated using permutation importance on the final trained models to identify which predictors contributed most to tumor volume predictions. In this approach, each feature was randomly permuted while holding all others constant, and the resulting decrease in model performance (measured by R^2) was recorded. Features that caused the largest drop in performance when shuffled were considered the most informative. This method provided an interpretable estimate of how strongly each temporal or volumetric parameter influenced the prediction outcome.

	feature	importance_mean	importance_std
5	log_ratio_1	1.130865	0.150734
1	Lag_1_mm3	0.984064	0.151392
2	Lag_2_mm3	0.003180	0.001573
6	log_ratio_2	0.001360	0.001321
0	Week_num	-0.000042	0.001757
4	dWeek_2	-0.001018	0.000518
3	dWeek_1	-0.003954	0.001718

1A. Random Forest feature importance.

	feature	importance_mean	importance_std
5	log_ratio_1	1.530379	0.295730
1	Lag_1_mm3	1.223347	0.209321
0	Week_num	0.038211	0.021671
6	log_ratio_2	0.003110	0.004478
4	dWeek_2	0.001878	0.000802
3	dWeek_1	-0.000616	0.001546
2	Lag_2_mm3	-0.002147	0.001373

1B. XG Boost feature importance

Figure 1. Example of Feature Importance Permutations. Shows the performance of the features and their contributions to predictions.

MODEL ROBUSTNESS

Parameters

Several key parameters were defined to optimize model performance and maintain consistency across patients. Kmin (4): the minimum number of weekly data points required per patient for inclusion in the model. N_estimators (800): the number of decision trees in the Random Forest and XGBoost ensemble. Test size (0.2): the proportion of the dataset reserved for model testing versus training. CV_splits (5): the number of folds used for group-based cross-validation, ensuring that entire patient datasets remained isolated between training and validation sets.

To ensure model robustness and generalizability across patients, a group-based cross-validation strategy was employed. Specifically, Group K-Fold cross-validation (GroupKFold) was used, in which the data were split into k distinct folds based on patient identifiers rather than individual data points. All train-tests splits were performed using patient level grouping, ensuring that each patient appeared exclusively in either the training or testing groups. This prevents the models from learning patient-specific patterns that would potentially increase performance on unseen data. Each individual time point was encoded with a value that corresponded to its patient grouping. This is to ensure longitudinal data was grouped together (i.e. Patient 01 time points = 0, Patient 02 time points = 1). Then using group

shuffle split, the patients were randomized and split into a test and training dataset based on patient encoding. This approach prevents data leakage between training and testing sets by ensuring that all samples from a single patient appear exclusively in one fold. This method allows for an unbiased assessment of the model's ability to predict tumor volume assessment for unseen (test) patients.

	fold	RF_MSE	RF_RMSE	RF_R2
0	1	202.259963	14.221813	0.887753
1	2	185.723257	13.628032	0.916490
2	3	484.218702	22.004970	0.867815
3	4	407.998757	20.198979	0.830541
4	5	1067.223484	32.668387	0.710061
5	avg	469.484833	20.544436	0.842532

Figure 2A. Random Forest Cross Validation

6	1	121.144653	11.006573	0.932769
7	2	116.321700	10.785254	0.947696
8	3	382.293920	19.552338	0.895639
9	4	225.160089	15.005335	0.906482
10	5	603.289418	24.561951	0.836101

Figure 2B. XGBoost Cross Validation

11	1	592.975381	24.351086	0.670920
12	2	1625.747943	40.320565	0.268985
13	3	767.292820	27.700051	0.790540
14	4	301.078629	17.351617	0.874949
15	5	1462.270967	38.239652	0.602737

Figure 2C. Neural Network Cross Validation

Figure 2. Example of Group KFold Cross Validation. Shows calculations of R squared values, root mean squared error, and mean squared error.

Overfitting

Tests were performed to identify whether the models generalized appropriately across patients and didn't just memorize individual tumor trajectories. Several overfitting mitigation strategies were employed throughout the modeling pipeline. A primary source of overfitting in longitudinal datasets is leakage across patients. To address this, the training and test datasets were checked to make sure that there was no overlap in data points between the splits. By searching for equivalent rows of data, and equivalent patients, It was determined that there had been no mixing of data points at any point throughout model building and fitting.

Regularization in Tree-Based Models

Both Random Forest and XGBoost models incorporated structural regularization to reduce model variance

Random Forest

- Limiting tree depth
- Increasing minimum samples per split and leaf
- Using a large number of trees to average over variability

XGBoost

- Shallow trees (max_depth = 2,3)
- Low learning rate(0.02)
- Subsample and colsample_bytree values <1.0

These constraints allowed for reduced model complexity and prevented the models from fitting noise in the training data

Additionally, to detect latent leakage or memorization, a feature-shuffling test was performed in which test-set features were randomly permuted. For a properly regularized model, such shuffling should destroy predictive performance. The resulting negative R2 values (random forests = -0.75, XGBoost = -0.69, NN = -0.7826) confirmed that no hidden leakage inflated model performance.

III. Results

Of the three models that were trained in this project, two produced promising prediction performance per this publication. The models trained using a Random Forest regression and XGBoost produced R^2 values of over 0.84 on testing data, and successfully predicted the overall RANO score for [RANO %] of patients. The neural network trained in this project did not perform as well as the other two models, with a R^2 value of 0.70. The poor performance of the neural network compared to the other machine learning models may be due to relative author inexperience in training methods and neuron design rather than incompatibility with the problem statement, as neural networks have been shown to predict tumor growth with high accuracy in other publications [12], [13]. XGBoost performed the best out of the three models, with an R^2 value of 0.90 on test data. However, all three models showed signs of overfitting despite attempts to mitigate this, and additional work is required to achieve clinical relevance at this time.

RANDOM FOREST

After training the Random Forest model, the R^2 value produced using the test data was 0.84. This indicates that it is a model that can accurately predict tumor sizes of patients found in this dataset but is somewhat overfitted, as the R^2 value obtained during training was 0.92. In Figure 3, the model can be seen to under-predict the size of tumors above 150 cm^3 when compared to the actual tumor size. This may be due to large tumors growing more erratically than smaller tumors from natural factors [11]. Predicted patient data from the testing portion of the dataset, Figures 4, 5, and 6 show actual vs predicted tumor volumes for 3 patients.

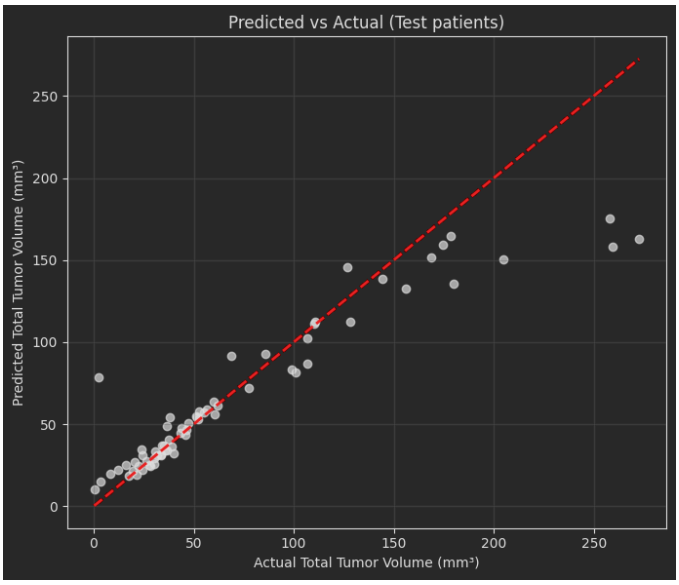


Figure 3. Predicted total tumor volume vs actual total tumor volume for Random Forest model in cm^3, showing a negative skew above 150 cm^3 of actual total tumor volume.

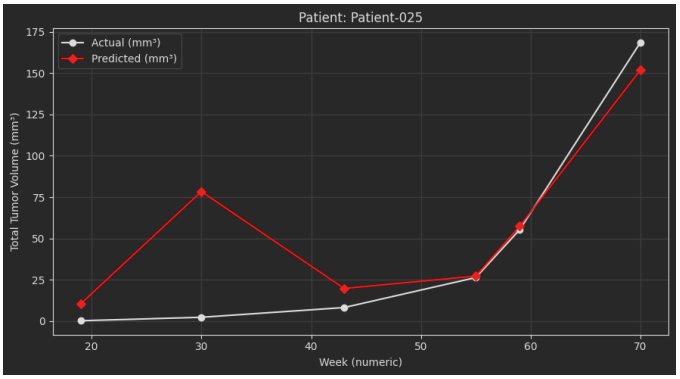


Figure 4. Predicted total tumor volume vs actual total tumor volume for patient 25 from Random Forest model in cm^3

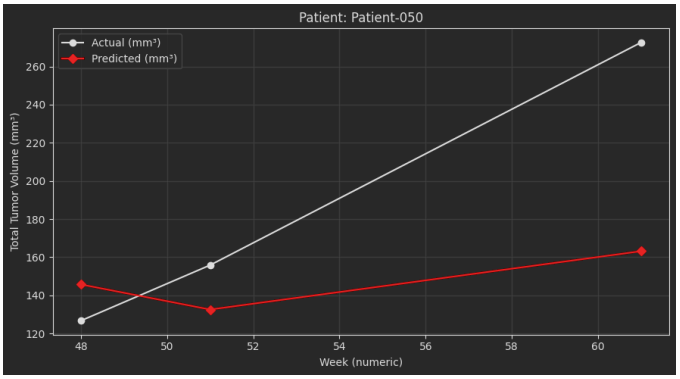


Figure 5. Predicted total tumor volume vs actual total tumor volume for patient 50 from Random Forest model in cm^3

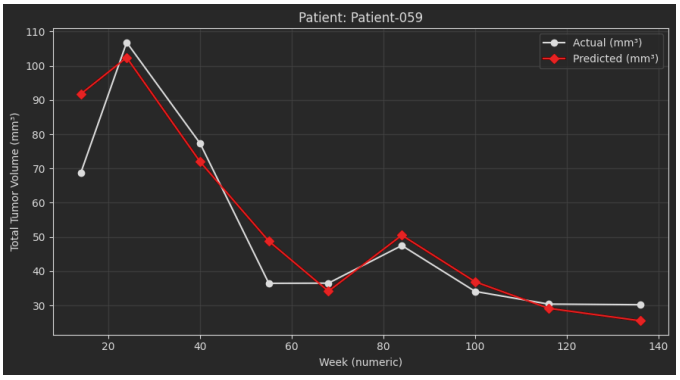


Figure 6. Predicted total tumor volume vs actual total tumor volume for patient 59 from Random Forest model in cm^3

XGBOOST

For the XGBoost model, the R^2 value produced using the test data was 0.88. This indicates that this model may be more accurate than the Random Forest model, however it suffers from the same limitation of consistently under-predicting large (150 cm^3 or above) tumors as seen in Figure 7.

Predicted patient data from the testing portion of the dataset is shown for the same patients as for the Random Forest model in Figures 8, 9, and 10, displaying the differences between predictions of the three models.

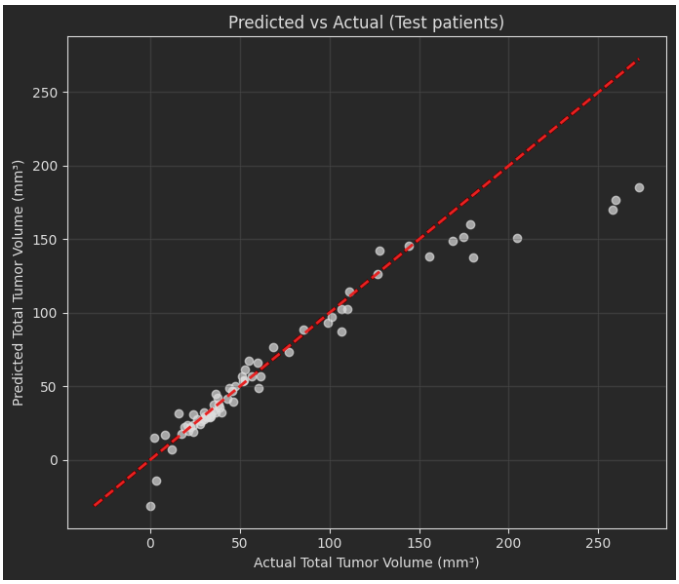


Figure 7. Predicted total tumor volume vs actual total tumor volume for XGBoost model in cm^3, showing a negative skew above 150cm^3 and below 5 cm^3 of actual total tumor volume.

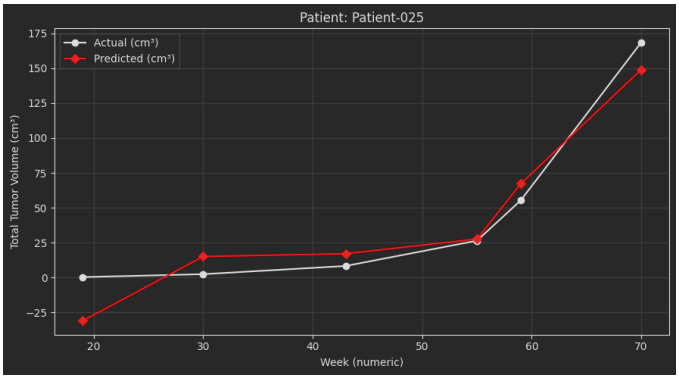


Figure 8. Predicted total tumor volume vs actual total tumor volume for patient 25 from XGBoost model in cm^3

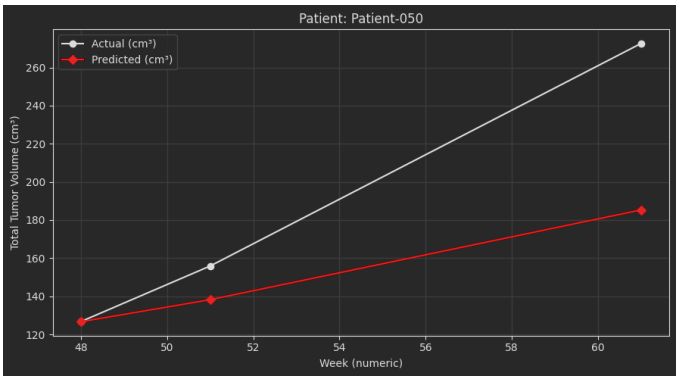


Figure 9. Predicted total tumor volume vs actual total tumor volume for patient 50 from XGBoost model in cm^3

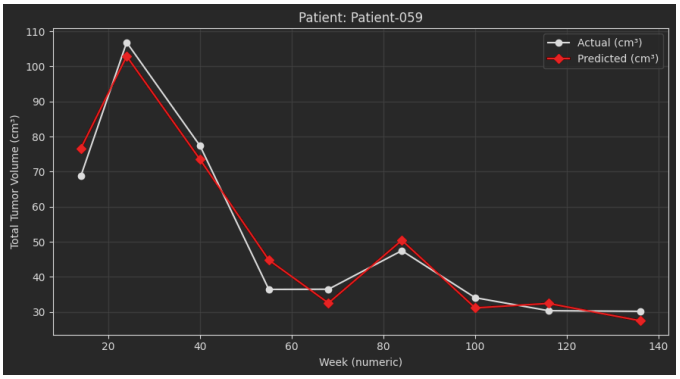


Figure 10. Predicted total tumor volume vs actual total tumor volume for patient 59 from XGBoost model in cm^3

COMPUTATIONAL NEURAL NETWORK (CNN)

The computational neural network generated a final R^2 value of 0.70 and had a more irregular plot of actual vs predicted values than the two machine learning models, Figure 5. Despite this comparatively poor performance, the predicted patient data from the same patients as previously shown manages to conform to the actual patient data fairly closely, Figures 12, 13 and 14.

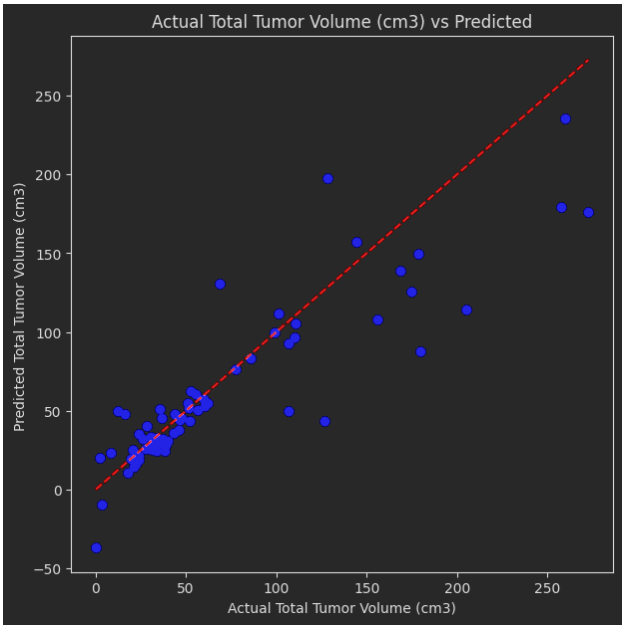


Figure 11. Predicted total tumor volume vs actual total tumor volume for Neural Network model in cm^3, showing quality loss above $75cm^3$ of actual total tumor volume.

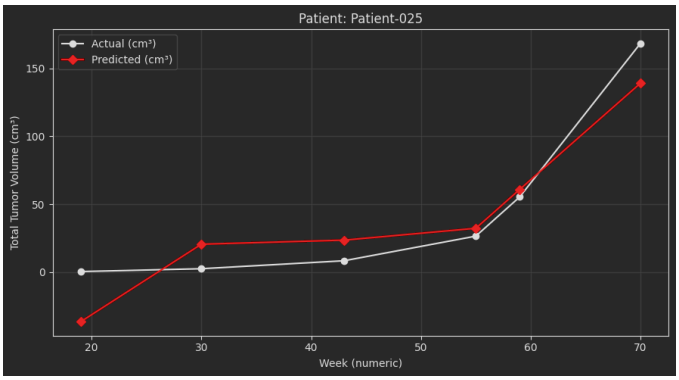


Figure 12. Predicted total tumor volume vs actual total tumor volume for patient 25 from Neural Network model in cm^3

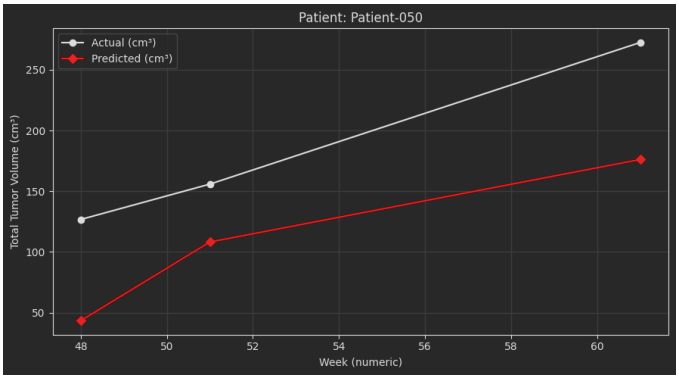


Figure 13. Predicted total tumor volume vs actual total tumor volume for patient 50 from Neural Network model in cm^3

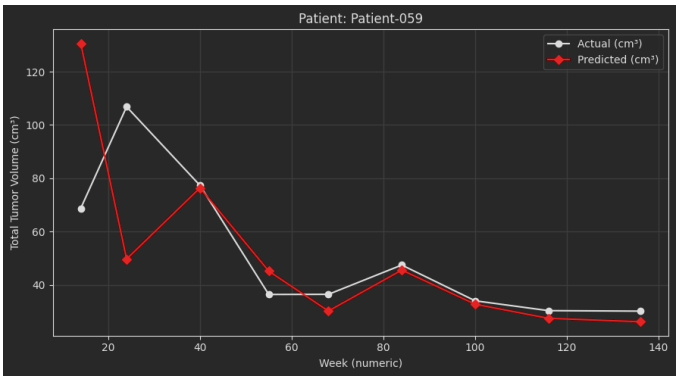


Figure 14. Predicted total tumor volume vs actual total tumor volume for patient 50 from Neural Network model in cm^3

IV. DISCUSSION

This study demonstrates that when supported by a robust longitudinal dataset as LUMIERE [1], temporally informed machine learning models can effectively capture glioblastoma progression. One of the main takeaways is that longitudinal MRI data enables predictive modeling that is not possible with

single-timepoint datasets, which form the basis of most GBM machine learning research [1], [6]. By including engineered temporal features such as the lagged tumor volumes, irregular imaging intervals, and logarithmic growth ratios, the models were able to learn clinically meaningful tumor growth trajectories rather than relying on just static snapshots. The shift towards modeling GMN as a dynamic process, which is an approach that aligns more closely with how clinicians monitor patients in real time [4].

Across the three models evaluated, Random Forest and XGBoost consistently achieved high predictive accuracy, with R^2 values ranging from 0.84 to 0.88 on patient's test data. The results suggest that ensemble tree-based methods are particularly well suited for longitudinal GBM prediction tasks, which are most likely due to their robustness to noise, their ability to handle nonlinear relationships, and their capacity to integrate different temporal features [5], [8]. Both of the models captured general tumor growth patterns and were able to reconstruct patient-specific trajectories, which is crucial for forecasting progression and potential recurrence. Although the neural network underperformed relative to the tree-based models, its somewhat accurate R^2 (=0.70) still demonstrated the feasibility of applying deep learning architectures to temporal GBM data [6], [8]. However, the model's lower performance does highlight the need for larger datasets or a more specialized architecture, for example RNNs or Transformers, that are designed explicitly for sequential data.

Another important takeaway concerns model behavior on edge cases. In patients with very large tumor volumes, or highly irregular growth, predictions tended to underestimate the actual volume. This pattern likely reflects biological variability, that is large tumors exhibit more unstable growth, and the limited representation of extreme outliers in the dataset [1], [9]. Similarly, patients with very few post-operative scans can pose challenges for all models, reinforcing that temporal ML methods require a minimum density of timepoints to meaningfully infer growth trends [1], [8]. These observations do highlight areas where future dataset expansion could improve predictive performance.

In cases where limited initial MRI data is available, for example with patient 50 (Figures 7, 10, and 13) where only 3 post-op MRIs were taken over a 13 week period, all three models struggled to accurately predict tumor growth. All models underpredicted tumor size at all but the initial time point, and all models underpredicted the final time point by over 80 cm^3. This may be caused in part by the data preprocessing performed, as only patients with a minimum of 5 time points following their initial surgery were used to train the models. As the models were trained using features including a 2 week lagging tumor volume change, it is not possible to reduce the minimum time point

requirement without severely limiting the functionality of the model. This indicates that a supplemental dataset with closer spaced imaging events may be beneficial in increasing the accuracy of the models at rapid changes, as that is when this project would be most beneficial in a clinical application [14].

The majority of the patients in this dataset experienced a progressive disease where their tumor volumes continually increased over time, or experienced periods of stability before their disease recurred with rapid growth in a short period of time such as that seen with patient 25 in Figures 6, 9 and 12. This allows the different models a large pool of training data to build off of and produce accurate results. However, as seen in Figures 8, 11, and 14 for patient 59, all three models can model decreases in tumor size with moderate accuracy. As the models are not grossly overpredicting the tumor volumes, it can be inferred that they have learned that tumor volumes can decrease, not only increase.

From a methodological perspective, the study's contribution includes a systematic implementation of patient-level group splitting to prevent leakage between training and testing sets. This is a critical but often overlooked challenge in longitudinal medical datasets. By ensuring that no patient appeared in both sets, the study provides a more realistic assessment of how well the models generalize to completely new patients, a requirement for any clinically deployable system [8], [10]. Additional evaluations, such as permutation based feature shuffling, confirmed that the models were not simply memorizing trajectories but were learning meaningful temporal relationships.

Model overfitting was a challenge in this project; despite taking steps to reduce overfitting such as limiting the number of trees in the machine learning model and poorly creating the neural network model, the models continued to predict the test data with high accuracy. In future modeling efforts, more emphasis should be placed on real world applications with completely novel data. These efforts may include sourcing a wider pool of data from which to train the models, combining multiple existing large datasets, such as from the Kaggle or Cochrane databases alongside the LUMIERE dataset.

Overall, our project contributes a novel modeling framework for predicting GBM progression using multi-timepoint MRI data and establishes baseline performance for several machine learning approaches applied to the LUMIERE dataset. It demonstrates that longitudinally engineered features substantially enhance predictive accuracy, and it highlights the clinical potential of using ML to forecast tumor growth without really relying on mechanistic models that require difficult-to-fit biological parameters [9]. These findings create a strong foundation for future work integrating radiomic

features, clinical covariates, and multimodal MRI, and they underscore the promise of machine learning as a supportive tool for personalized GBM monitoring and treatment planning [1], [5], [6].

V. Conclusions

This research highlights the importance of utilizing MRI data to model glioblastoma (GBM) development and forecast patient-specific tumor volume fluctuations over time. Using the LUMIERE dataset, which offers -timepoint postoperative imaging for 91 patients [1], we created machine learning models that effectively capture clinically relevant tumor-growth patterns. Of the three regression models evaluated, Random Forest and XGBoost displayed the performance with R^2 scores ranging from 0.84 to 0.88 indicating that tree-based ensemble approaches are highly successful, for temporal tumor forecasting [5], [8]. Our approach to feature engineering, including delayed tumor volumes, uneven imaging timing and logarithmic growth metrics was crucial, for capturing growth patterns and enhancing prediction precision. These efforts emphasize the promise of data-centric techniques to advance from fixed imaging evaluation to prediction of GBM development [6].

Although these promising outcomes are evident, multiple directions remain for research. A logical progression involves broadening the feature set by including MRI modalities like FLAIR or T2-weighted scans alongside radiomic features that describe tumor morphology, texture and contrast enhancement characteristics [1]. Adding factors, such as patient age, therapeutic protocols or molecular indicators such as MGMT and IDH status, could enhance personalized prediction accuracy. From a standpoint sequential deep learning models like LSTMs, GRUs or Transformers might be investigated to more effectively capture long-term temporal relationships, particularly with the availability of larger longitudinal datasets [8]. Ultimately broadening the scope, beyond the LUMIERE dataset by integrating longitudinal GBM datasets would aid in confirming generalizability and advance towards clinical application [1], [7].

Overall, this project provides an initial but meaningful step toward predictive modeling of GBM evolution. By demonstrating that machine learning can track and forecast tumor growth using routinely collected MRI data, this work lays the groundwork for future tools that could support more personalized monitoring and treatment planning for GBM patients [4], [6].

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